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Introduction to Machine Learning

Project Report

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1. Introduction

Voice-based systems are becoming more common in everyday technology, as they allow interaction using only a microphone and spoken commands. Such solutions are used in many applications, including security and access-control systems.

In this project, machine learning methods are used to study how voice recordings can be analyzed to support access decisions. The goal is to build and evaluate a complete recognition pipeline and examine how different design choices influence the final system performance.

The next sections describe the task formulation, dataset properties, and the methods applied during training and evaluation.

2. Dataset and Task Overview

2.1 Problem Definition

The objective of this project is to design a voice-based recognition system capable of distinguishing between authorized and unauthorized speakers. The task is formulated as a binary classification problem, where each input sample corresponds to a short voice recording represented in the form of a spectrogram.

The classification output determines whether access should be granted or denied. Due to the security-oriented nature of the problem, the system must balance robustness and reliability, rather than focusing solely on overall accuracy.

2.2 Dataset Characteristics

The dataset used in this project contains voice recordings originating from multiple speakers and recording conditions. The data includes variability in speaking style, intonation, recording quality, and environmental noise. Such diversity is essential for evaluating the generalization ability of the trained models.

Because raw audio signals differ in duration and acoustic properties, a dedicated preprocessing pipeline is required before model training. This pipeline includes data cleaning, segmentation, feature extraction, and augmentation procedures. The next section presents the data augmentation methods used in this project and explains how they contribute to improving model robustness.

3. Data Augmentation

Prior to training the convolutional neural network (CNN), we constructed a custom voice dataset and applied a series of data augmentation techniques.

The dataset consists of full-length voice recordings collected from publicly available online sources, such as video essays, audiobooks, and speeches, as well as original recordings created by the project participants.

3.1 Raw Data Collection

As mentioned in the previous section, audio samples were collected from multiple sources and across different content genres. The recordings can be broadly divided into two main categories.

Publicly available audio data.

The first category consists of audio files obtained from the internet, including audiobooks, speeches, and video essays. These sources were selected because they are readily available and typically contain a moderate level of background noise, such as recording artifacts or environmental sounds. As a result, they provide realistic voice data that better reflects real-world recording conditions.

Participant-recorded audio data.

The second category includes audio recordings created by the project participants themselves. Each participant recorded multiple voice samples, with total recording durations ranging from approximately 15 to 20 minutes per participant. This data was important for ensuring controlled and consistent speech content, as well as for supplementing publicly sourced recordings with samples recorded under known conditions.

3.2. Preprocessing Pipeline

We implemented a custom Python pipeline using the `librosa` and `soundfile` libraries to convert raw audio into standardized inputs for the CNN.

Segmentation

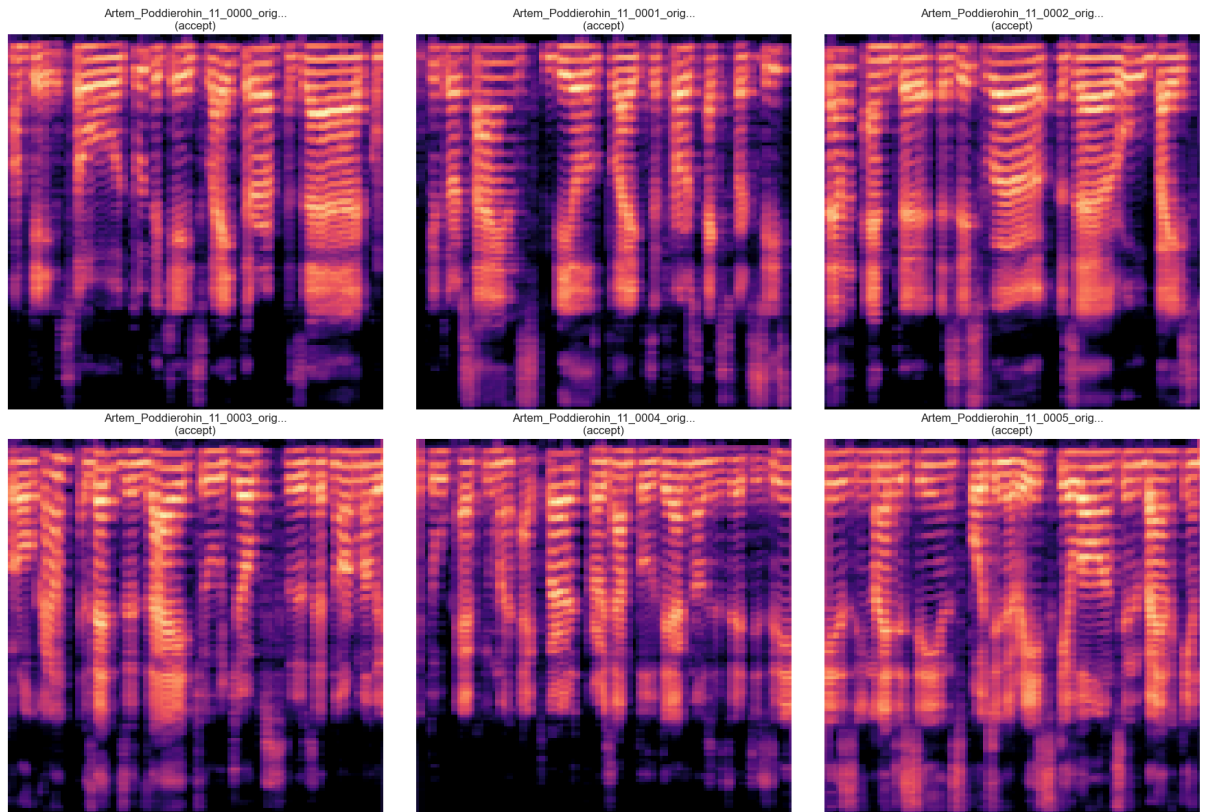
Long recordings were split into fixed-length segments (defaulting to 3.0 seconds). Segments shorter than the target length were discarded to ensure uniform data quality.

Feature Extraction

We converted audio segments into Mel spectrograms. We used a standard FFT window size (`n_fft=2048`) and hop length (`hop_length=512`) with 128 Mel bands. The resulting power spectrograms were converted to the decibel (dB) scale (`librosa.power_to_db`).

Output Formatting

Spectrograms were saved as 128x128 pixel PNG images. This was achieved by setting the Matplotlib figure size to 1.28x1.28 inches with a DPI of 100, ensuring the input dimension matches the network's expected input size.

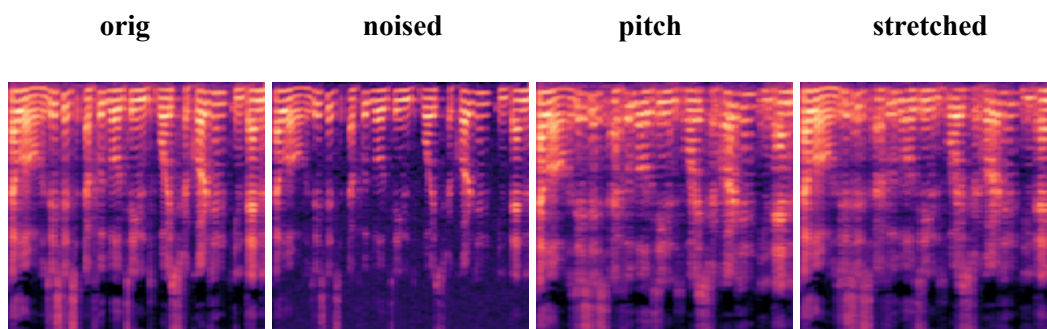


3.3. Augmentation

In order to meet the project's robustness requirements, we applied a flexible augmentation registry allowing for both online and offline transformations:

Noise Injection: We added Gaussian noise with specific intensity levels (e.g., 0.005) to simulate environmental interference.

Pitch Shifting & Time Stretching: We utilized `librosa.effects` to shift the pitch (`n_steps`) and stretch time (`rate=1.1`), effectively multiplying the dataset size and forcing the model to learn invariant features rather than memorizing specific frequencies or speeds.



4. Exploratory Data Analysis

To understand the characteristics of our voice dataset and verify the quality of our preprocessing, we conducted a comprehensive Exploratory Data Analysis (EDA) using `librosa` for spectral analysis and `parselmouth` for voice quality metrics.

4.1. Dataset Distribution

The dataset is composed of distinct "Authorized" (Class 1) and "Unauthorized" (Class 0) identities.

Class 1 (Authorized): Consists of original recordings from project participants (e.g., *Artem*, *Dmitry*, *Valentyn*). These recordings typically range from 30 to 300 seconds before segmentation.

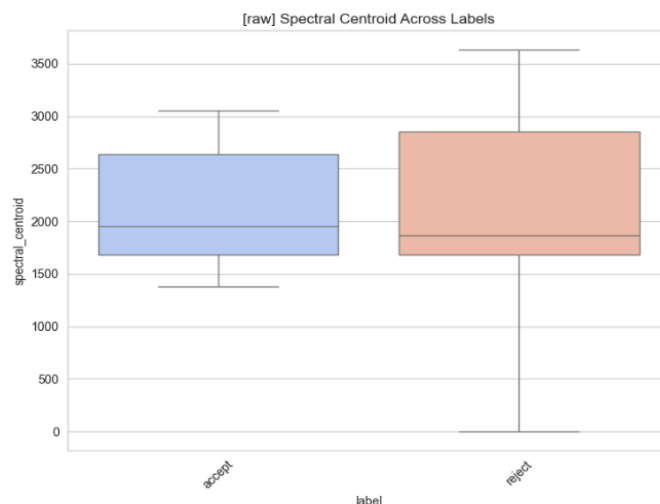
Class 0 (Unauthorized): Consists of diverse public audio sources, including political speeches (*Obama*), audiobooks (*The Egg*, *Door in the Wall*), and celebrity interviews (*Leonardo DiCaprio*).

We verified the balance between classes after segmentation. While raw file durations vary (e.g., some audiobooks are >10 minutes), the segmentation into 3-second chunks ensures a more balanced distribution of training samples.

4.2. Spectral Analysis

We examined the spectral characteristics of the "Authorized" vs. "Unauthorized" voices to identify distinguishing features.

Spectrogram Comparison:

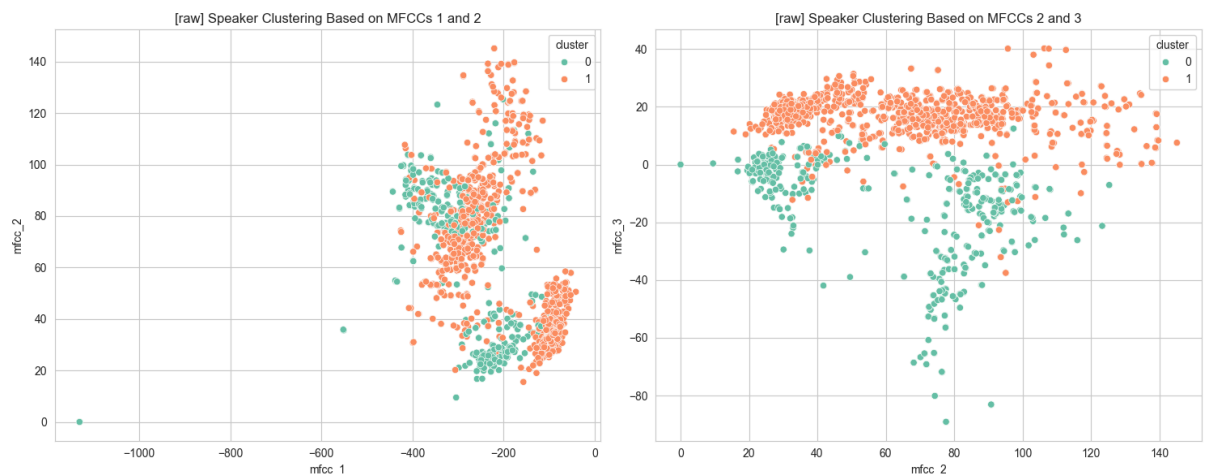


Visual inspection of Mel-spectrograms reveals that "Authorized" recordings (typically recorded in quiet home environments) often have distinct harmonic structures compared to "Unauthorized" samples, which may contain different background noise profiles or recording signatures.

4.3. Feature Space and Separability

To assess the feasibility of the classification task, we extracted high-level features such as MFCCs (Mel-frequency cepstral coefficients).

Dimensionality Reduction: We applied Principal Component Analysis (PCA) to project these high-dimensional features into 2D space.



Principal Component Analysis (PCA) reveals that the first two components, which capture the highest variance, do not distinctively separate the classes. This suggests that the most dominant variations in the dataset are likely generic acoustic features common to all recordings, such as amplitude or background noise levels.

However, plotting PC2 and PC3 reveals a much clearer separation between 'Authorized' and 'Unauthorized' samples. This indicates that the distinguishing vocal characteristics (timbre, pitch, harmonic structure) are present and separable but exist as latent features rather than dominant ones. This finding supports the choice of a Convolutional Neural Network (CNN) over simpler linear classifiers, as the CNN can effectively filter out the dominant 'noise' (PC1/PC2) and focus on the subtle, class-specific features found in higher-order components.

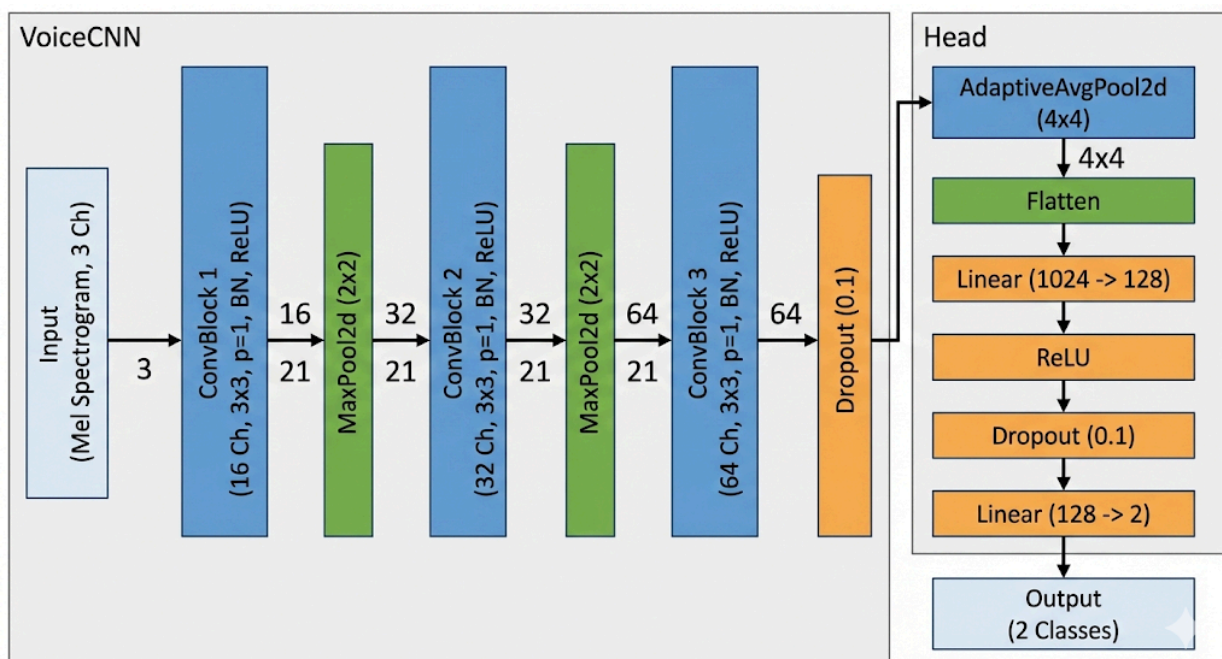
5. Model architecture

To work within a voice recognition domain, we designed a custom Convolutional Neural Network (VoiceCNN). This custom model is designed to work with 2D spectrogram images, thus transferring the voice recognition task to the image recognition domain.

5.1. Structural Design

The model follows a hierarchical structure consisting of feature extraction layers followed by a classification head. It includes:

- **Convolutional Backbone:** The architecture utilizes three primary ConvolutionBlock modules with increasing filter depths. Each block performs a 2D convolution with a 3X3 kernel and same padding to preserve spatial dimensions.
- **Batch Normalization:** Batch Normalization is applied directly after the convolution but before the activation function. This stabilizes the learning process and allows for higher learning rates.
- **Max-pooling:** Max-pooling layers with a 2X2 window and a stride of 2 are located between the convolutional blocks. This reduces the spatial resolution of the feature maps, and increases the receptive field and reduces computational complexity.
- **Classification Head:** To ensure compatibility with input dimensions, we employ an Adaptive Average Pooling layer. It forces the output to a fixed 4X4 spatial size. This is followed by a flattened dense layer of 128 neurons and a final linear layer for binary classification (Authorized vs. Unauthorized).



5.2. Activation, Regularization and Optimization

Activation Functions: While the architecture supports multiple types, the best-performing models utilize ReLU (Rectified Linear Unit) for hidden layers, providing a sparse representation and mitigating the vanishing gradient problem.

Dropout: Dropout layers are placed after the third convolutional block and within the dense head. For our top-performing configurations, a dropout rate of 0.1 proved most effective in preventing the model from over-relying on specific spectrogram artifacts.

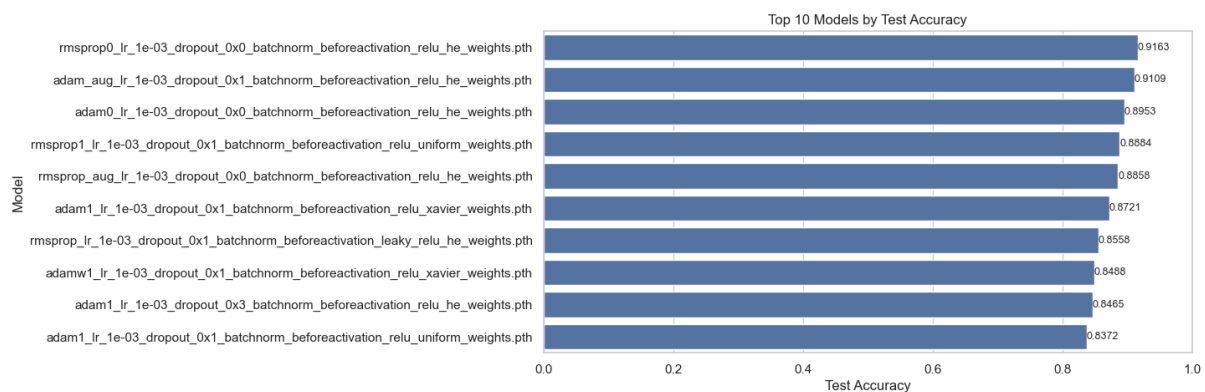
Weight Initialization: The model uses He (Kaiming) Normal Initialization, which is mathematically optimized for layers followed by ReLU activations, ensuring that signal variance remains consistent across deep layers.

Optimizers: The final model performance is heavily influenced by the choice of optimizer and its hyperparameters. Based on our experimental results, the best performing were two configurations: RMSprop and Adam.

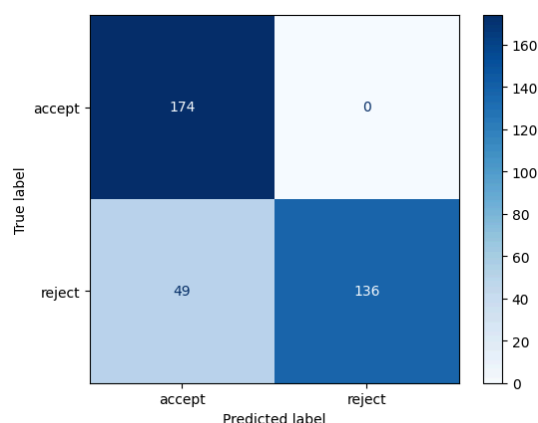
6. Evaluation

6.1 Model Comparison and Results

We trained multiple variations of the model to test the impact of different architectural choices. The graph below summarizes the performance of our top configurations on the validation set.



Our experiments confirmed that placing Batch Normalization before the ReLU activation significantly stabilized training compared to the baseline. The addition of He Initialization and Dropout provided the best balance, reducing the False Acceptance Ratio. The confusion matrix below details the final decision boundaries.



7. Discussions

7.1 Requirments

- ☒ data augmentation
- ☒ choosing the length of a single excerpt, input normalization
- ☒ changing the number and size of the layers
- ☒ different optimizers (SGD, Adam), learning rate schedule, weight decay
- ☒ batch normalization
- ☒ dropout
- ☒ Monte Carlo dropout for uncertainty measurement
- ☒ activation functions tested against different weight
- ☒ initializations

7.2 Key Findings:

Architectural Efficiency: Our custom 3-block CNN architecture proved sufficient for this task. We found that specific architectural choices were critical for stability. Notably, placing Batch Normalization before activation combined with He Initialization yielded the most consistent convergence and highest validation accuracy.

Data Robustness: The Exploratory Data Analysis (EDA) revealed that while simple features (like volume) were shared across classes, the latent vocal characteristics were distinct. Our augmentation pipeline successfully forced the model to learn these robust features rather than overfitting to clean, studio-like conditions.

Feasibility: The final model achieved a balanced trade-off between the False Acceptance Ratio and False Rejection Ratio. This balance is crucial for a security system, ensuring that authorized users are not locked out (low FRR) while strictly preventing unauthorized access (low FAR).

8. Conclusion

In this project, we successfully developed and evaluated a deep learning module for a voice-based intercom access control system. By treating voice recognition as an image classification problem using Mel-spectrograms, we demonstrated that a custom Convolutional Neural Network (CNN) can effectively distinguish between authorized and unauthorized personnel. Future work could focus on expanding the "Authorized" dataset to include more diverse acoustic environments to further reduce the risk of microphone bias.